Homework 1

Autonomous Mobile Robots

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Instruction

This document provides the results of the codes of the assignment provided in the Codes folder, Please note that to run the codes all the codes have to be in the same directory and that for questions 2,3,4,5 there are two codes one that run the simulation without animation and provides the final result and and another one that provides the animation of the simulation required. Also please note that some of the initial configurations of the objects in the simulation have been hard coded instead of randomizing to produce results that lie in the workspace. You can alter these to see different behaviours.

Assignment Instructions

Implement a non holonomic vehicle in Matlab as explained in class with the following specs

- max steering angle = pi/4
- max speed = 5 m/s
- workspace dimension 200m x 200m. The global frame is at (0,0)
- create and represent a desired shape for your robot (rectangle, star,...) but not a triangle
- The robot's initial configuration is (100, 100, rand(θ))

Question 1

Implement a homogeneous transformation using a Matlab function and test it by translating the robot by (x=20, y=30) and rotating by ($\pi/6$). Plot the initial configuration and the final configuration in the workspace. You can modify and adapt the provided code

Solution

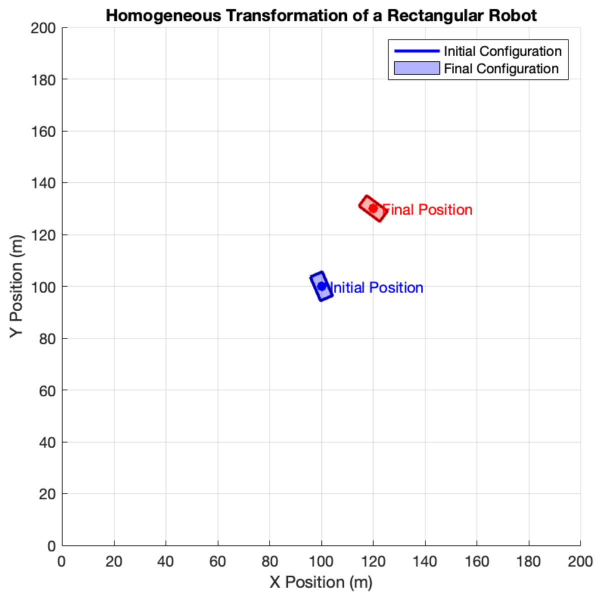

Figure 1: Homogeneous Transformation

Question 2

Go to goal. In this simulation pick a random goal point in the environment and drive the robot to the point as explained in class. The initial pose of the robot is as calculated in 1. The robot initial and final velocity have to be 0. Plot x and y position, and velocities of the robot with respect to time.

Solution

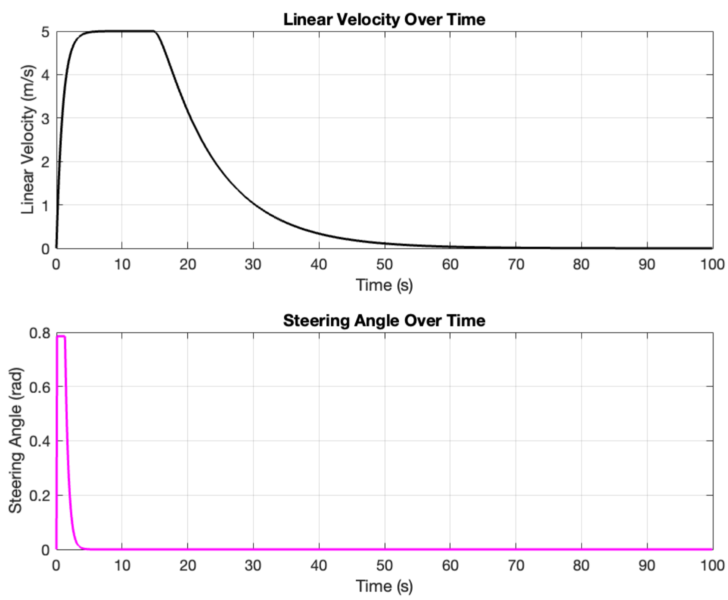

Figure 2: Velocity And Steering Angle Over Time

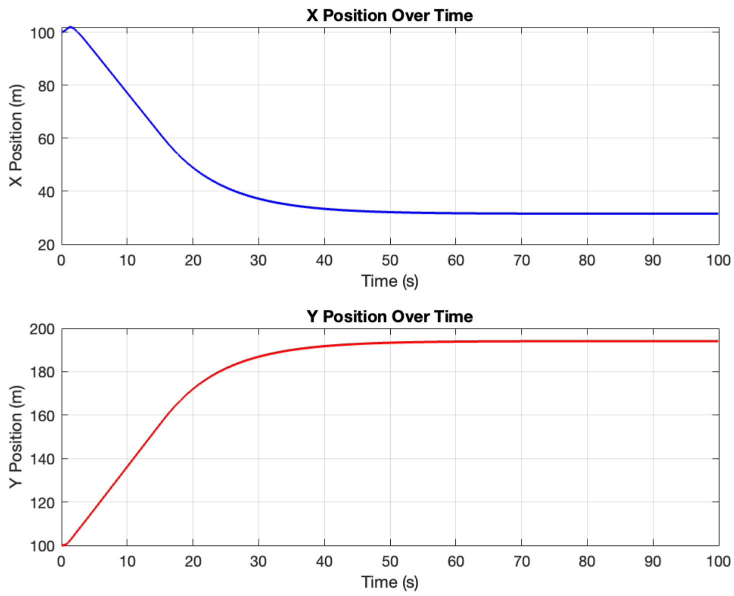

Figure 3: (x,y) Position Over Time

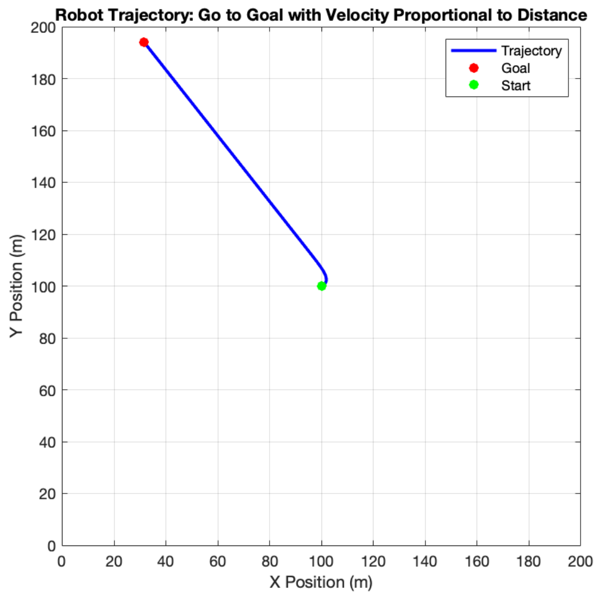

Figure 4: Simulated Path

Question 3

Implement a cruise controller (i.e., PID as seen in class) with vref = 3 m/s. Make the robot go through 3 random points in the environment without stopping switching to next goal once the current goal has been visited. Assume that the velocity has the following dynamics $v(t+dt) = v(t) + u(t) - 0.01v(t)$ where u is the output of the PID and dt=0.1. Plot the velocity of the robot.

Solution

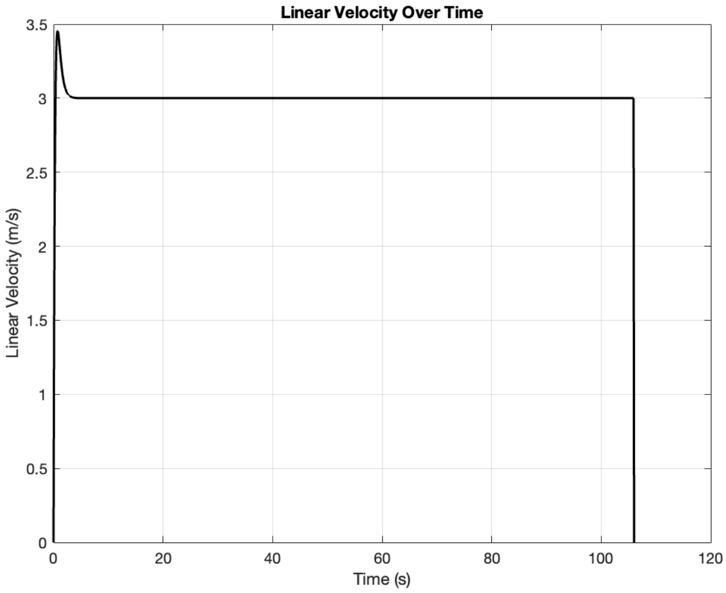

Figure 5: Linear Velocity Over Time

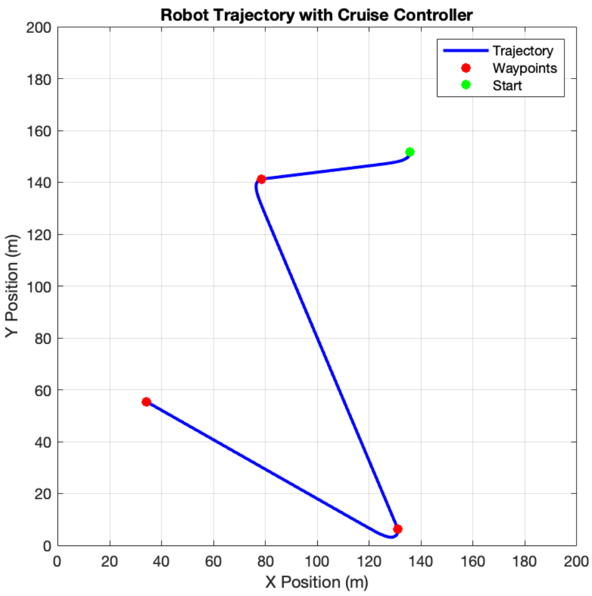

Figure 6: Simulated Path

Question 4

follow a wall: draw a line in the workspace and make the robot follow the line at a constant distance (10m) from the line, as explained in class. Plot the relative distance between robot and line over time.

Solution

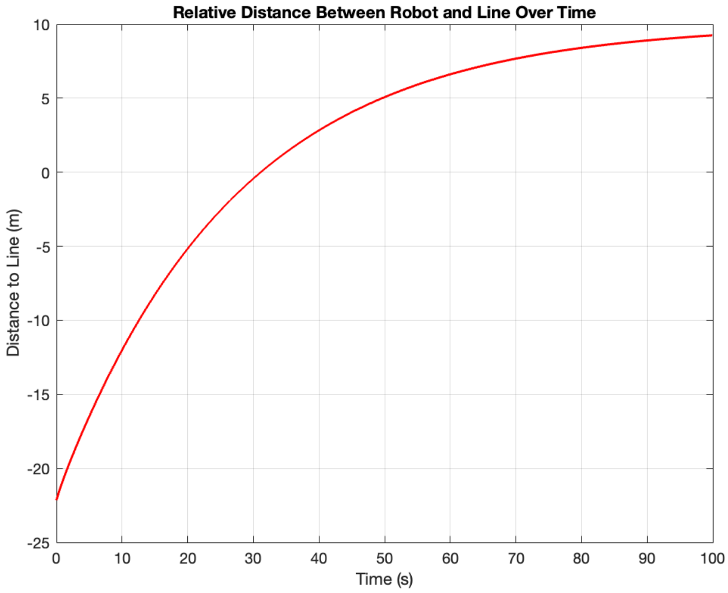

Figure 7: Relative Distance Between Robot and Line Over Time

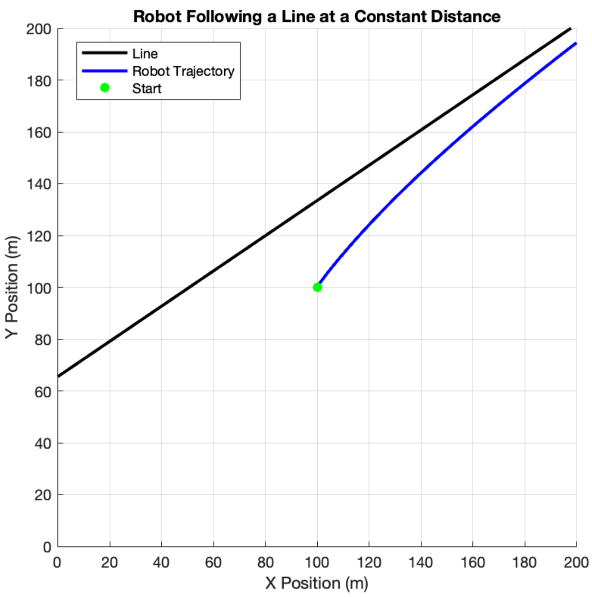

Figure 8: Simulated Path

Question 5

Follow a circle: draw a circle of radius r = 75m in the center of the workspace and make the robot follow it from any starting point of the workspace. Plot the position and velocities of the robots.

Solution

The simulated path shows some small circles occurring on the the path because the robot reaches the point because of the speed of the point and is forced to track back because it can't cross it. It can be better understood through running the animation code.

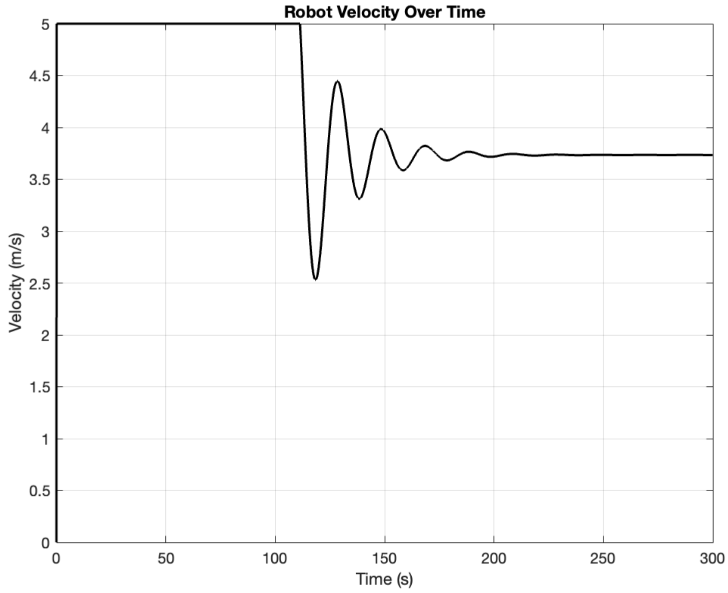

Figure 9: Robot Velocity Over Time

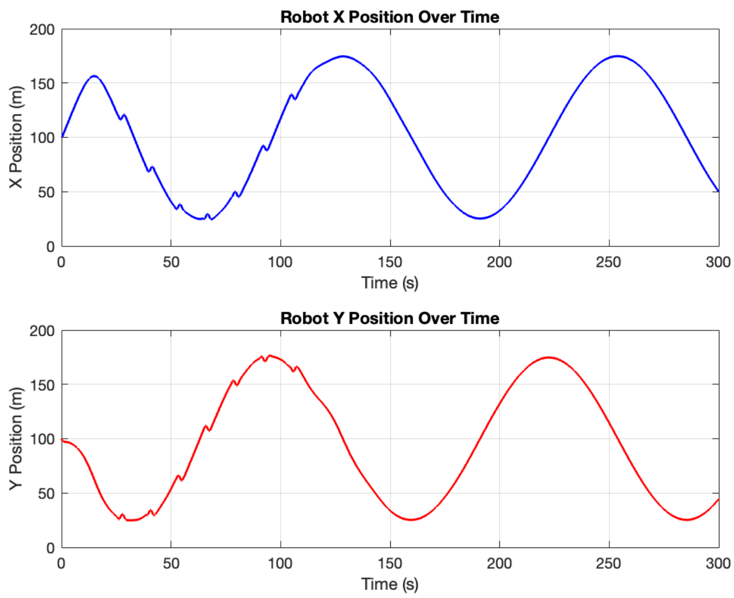

Figure 10: (x,y) Position Over Time

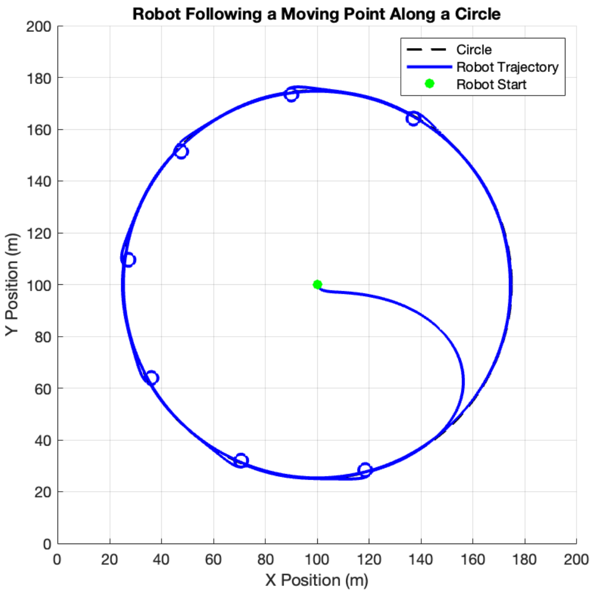

Figure 11: Simulated Path